M-Series Motors: Electrical Test Procedure

Restricted document. These tests use high DC voltages and live measurements. They may only be carried out by qualified electrical technicians authorised under the site electrical safety rules.

1. General requirements

- The motor must be isolated and locked out according to SOP-LOTO-001 for all tests except the live current measurement in section 5.
- Verify the absence of voltage at the motor terminals with a tested voltage detector before disconnecting the supply cables.
- Disconnect the supply cables and any surge capacitors, VFD outputs and PT100 sensor leads before applying a test voltage.
- Wear insulating gloves rated for the test voltage and keep other people out of the test area with barriers and a sign.

2. Insulation resistance test

Use an insulation tester (megger) with a test voltage of **1000 V DC** for motors rated 400 V.

1 Connect the tester between the three phases joined together and the motor frame (earth).

2 Apply the test voltage for **1 minute** and record the insulation resistance (IR) value at 1 minute.

3 Record the winding temperature. Correct the reading to 40 °C: the insulation resistance halves for every 10 °C increase in temperature.

Result (corrected to 40 °C)	Assessment
Above 100 MΩ	Good
10 to 100 MΩ	Acceptable; trend at next test
Below 10 MΩ	Do not energise. Dry the winding and re-test
Below 1 MΩ	Do not energise. Winding repair required

3. Polarization index (PI)

The polarization index is the ratio of the insulation resistance at 10 minutes to the resistance at 1 minute, measured with the same test voltage.

PI = IR(10 min) / IR(1 min)

PI	Assessment for Class F insulation
Above 2.0	Good
1.0 to 2.0	Questionable; moisture or contamination likely
Below 1.0	Do not energise

The PI test is not meaningful when the 1-minute insulation resistance is above 5000 MΩ.

3.1 Discharging the winding

After any insulation test the winding stores a dangerous charge. Short-circuit the winding to earth with the tester's discharge function or an earthing lead for **at least four times the test duration** (at least 40 minutes after a 10-minute PI test) before touching the terminals.

4. Winding resistance test

Measure the resistance of each phase with a low-resistance ohmmeter (4-wire method). Correct all values to the same temperature.

- The difference between the highest and lowest phase resistance must not exceed **2%** of the average.
- A higher difference indicates a shorted turn, a poor connection or a broken conductor.

5. Live current imbalance measurement

This measurement is made with the motor running and the terminal box covers closed where possible. Use a clamp meter rated CAT III 600 V or higher, measuring on the supply cables in the starter panel.

1 Measure the current in each of the three phases with the motor at normal load.

2 Calculate the current imbalance: the largest deviation from the average current, divided by the average current, times 100.

3 A current imbalance above **10%** requires investigation of the supply voltage balance, the connections and the winding.

6. Records

Record all results on the motor test sheet: date, motor ID, test voltage, IR at 1 and 10 minutes, PI, winding temperature, phase resistances and the name of the technician. Attach the sheet to the maintenance ticket. Compare results with the previous test: a fall in insulation resistance of more than 50% since the last test must be investigated even when the absolute value is acceptable.